

Optimal Longitudinal Stop Control System Considering Human Factor via Human-Robot Interaction for Personal Mobility Service Robots

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Abstract: In this paper, we propose an optimal longitudinal control system for personal mobility service robots, considering human factors derived from Human-Robot Interaction (HRI) experiments. First, we design velocity profiles for different modes using Bezier curves to improve the driving safety of personal mobility service robots. We analyze individual mode preferences through usability evaluation in HRI experiments and derive a human factor that reflects user needs. Based on the derived human factor, we construct an optimal velocity profile that ensures user preference and convenience. To accurately track the optimal velocity profile while ensuring both driving performance and user safety, we design a model-based optimal control system using a Linear Quadratic Regulator (LQR) integrated with an error-based compensator. The proposed optimal longitudinal control system, considering human factors, is implemented and validated on an autonomous electric wheelchair equipped with autonomous driving sensors, demonstrating its effectiveness in providing a control framework for personal mobility service robots.

Keywords: Personal Mobility Service Robot, Human-Robot Interaction, Human Factor, Optimal Longitudinal Control

1. INTRODUCTION

Recently, autonomous driving technology has rapidly advanced, extending beyond conventional vehicles to a broader range of mobility platforms [1]. As the personalized service industry grows, research and development efforts are increasingly focusing on mobility solutions equipped with autonomous driving capabilities. Representative examples of personalized mobility include autonomous electric wheelchairs, autonomous kickboards, autonomous bicycles systems. Among these, the autonomous electric wheelchair demonstrates strong potential as a primary mobility solution—not only for transportation-vulnerable groups such as the elderly, individuals with disabilities, and pregnant users, but also for the general public—owing to its superior safety and comfort in both indoor and outdoor environments compared to other autonomous mobility systems [2]. In fact, autonomous electric wheelchairs are already being deployed and operated in public facilities such as hospitals and airports [3].

Autonomous electric wheelchairs are evolving into the domain of Human-Robot Interaction (HRI), enhancing user convenience and accessibility requirements [2]. To provide safe and comfortable mobility, these systems enable personalized motion planning derived from user needs. In complex indoor and outdoor environments, the wheelchair distinguishes between drivable and hazardous areas, allowing for optimal path generation that improves

both user safety and efficiency [4]. Furthermore, accurate localization of obstacles enables collision risk assessment and proactive avoidance, enhancing user safety and meeting individual mobility needs [5]. Moreover, by analyzing user's real driving data and adapting to driving habits, personalized shared control systems can be implemented to improve driving comfort [6].

The user needs-based techniques for enhancing user safety and driving comfort are most effective when integrated with precise optimal control strategies. To achieve this, model-based optimal control strategies such as Model Predictive Control (MPC) have been applied for safe and precise operation of autonomous mobility platforms [7]. Furthermore, control parameters can be optimized using machine learning techniques to improve both ride comfort and driving safety [8]. Recent approaches include vision-based control systems that track user's eye movements, enabling user-centric control of electric wheelchairs [9]. In real-world environments, control systems must also cope with uncertainties, nonlinear dynamics, and external disturbances. To this end, disturbance compensation mechanisms have been developed to improve control performance under such conditions [10]. Path-tracking compensators have been designed to mitigate trajectory errors caused by uneven road surfaces [11]. Moreover, recent studies propose machine learning-based compensators that account for user stability during velocity control [12].

In this paper, we propose an optimal longitudinal ve-

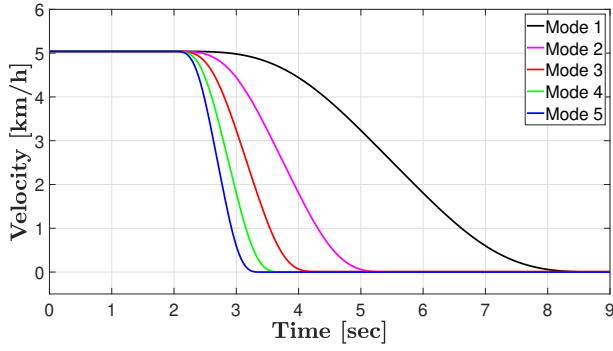


Fig. 1. Velocity profiles of five deceleration modes on HRI experiments.

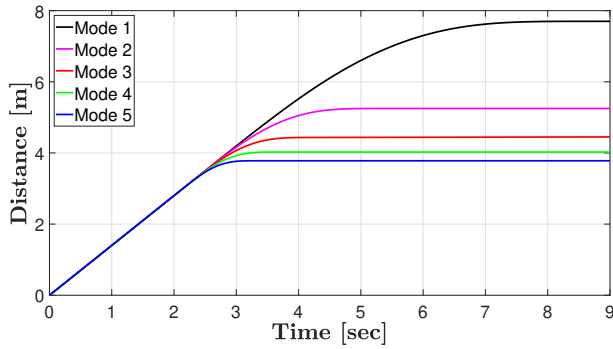


Fig. 2. Distances of five deceleration modes on HRI experiments.

locity control system for personal mobility service robots, such as autonomous electric wheelchairs, considering a human factor derived from HRI experiments. The proposed approach incorporates a Bezier curve-based velocity planning method to improve driving safety, where velocity profiles for different modes are designed. A human factor is then derived through HRI usability evaluations to construct an optimal velocity profile that reflects individual user preferences and enhances convenience. To accurately track the desired optimal velocity profile, we develop a model-based optimal control strategy grounded in the dynamics of the autonomous wheelchair. Additionally, an error-based compensator is integrated to reduce tracking deviations, thereby enhancing driving safety and comfort.

2. PROBLEM STATEMENT

Autonomous electric wheelchairs, a representative form of personal mobility service robots, require precise control to ensure both user safety and driving comfort. However, most existing systems rely on heuristic rules or fixed-gain controllers that apply uniform deceleration patterns [8, 13]. Since these approaches do not reflect individual user needs or ensure accurate driving performance, they fail to provide personalized control, thereby limiting their realization as practical personal mobility service robot solutions.

We approach this challenge through optimal control that considers the user's needs and improves driving per-

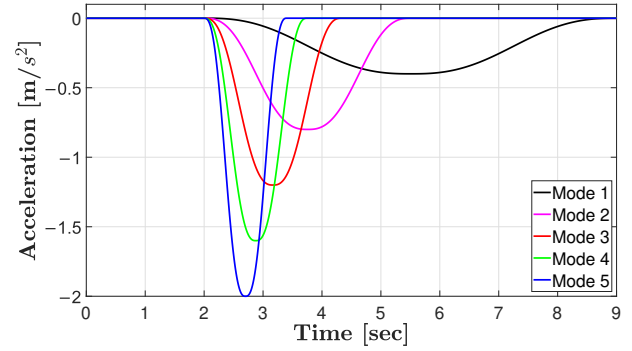


Fig. 3. Accelerations of five deceleration modes on HRI experiments.

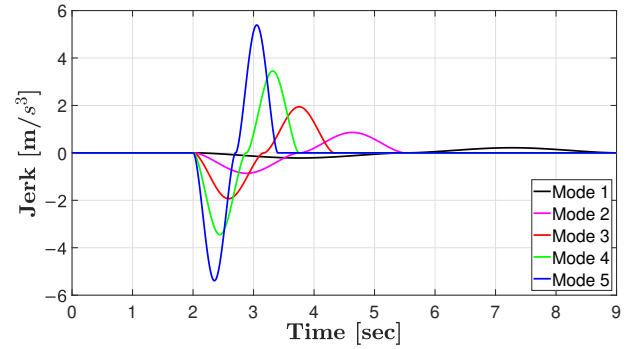


Fig. 4. Jerks of five deceleration modes on HRI experiments.

formance. In this study, we conduct human-robot interaction experiments to evaluate the usability of an autonomous electric wheelchair. Based on the results of the HRI experiments, we derive human factors to design a velocity profile that reflects the user's needs. Furthermore, we apply an optimal control system with an error-based compensator to obtain the driving performance of a personal mobility service robot.

3. HUMAN-ROBOT INTERACTION WITH PERSONAL MOBILITY SERVICE ROBOT

3.1 Velocity Profile Design for HRI Experiment

Usability evaluation of an autonomous electric wheelchair HRI experiment will be conducted for deceleration driving evaluation. In this experiment, participants are asked to ride the wheelchair at a constant speed of 5 km/h and experience five distinct deceleration modes. After each experience, participants identify the most preferred (Best) and least preferred (Worst) deceleration mode. To reflect a range of deceleration experiences, five velocity profiles design with average deceleration rates of -0.2 , -0.4 , -0.6 , -0.8 , and -1.0 m/s². The term "average deceleration" refers to the total change in velocity from the initial value to zero, divided by the elapsed time until the complete stop.

$$\mathcal{A} = \{-0.2, -0.4, -0.6, -0.8, -1.0\} \quad [\text{m/s}^2] \quad (1)$$

where $\mathcal{A}$ is the set of deceleration modes used for evaluat-

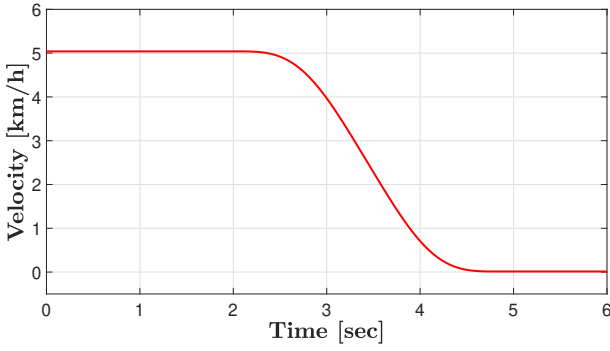


Fig. 5. Optimal velocity profile considering human factor.

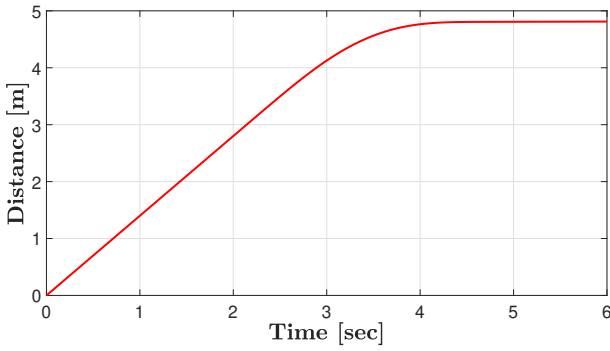


Fig. 6. Distance of optimal velocity profile considering human factor.

ing the user preference for deceleration. Each value corresponds to a deceleration mode tested in the HRI experiment and serves to generate Bezier-based velocity profiles with varying deceleration intensities.

The Bezier curve is a useful profile generation method due to its characteristic of ensuring continuous and smooth acceleration. This results in a more natural deceleration experience, effectively minimizing sudden changes in motion that could otherwise lead to passenger discomfort or mechanical disturbances such as wheel slip. This contributes to improved driving comfort and stability, which are critical in the design of personal mobility service robot systems [14].

$$a_{B,1}(t) = \sum_{i=0}^6 \binom{6}{i} (1-t)^{6-i} t^i P_i \quad (2)$$

where $a_{B,1}$ is a Bezier curve-based acceleration profile defined over the normalized time interval $t \in [0, 1]$. $\binom{6}{i}$ is the binomial coefficient, i is control point index. The profile is constructed as a 6th-order Bezier polynomial and P_i denotes seven control points in the two-dimensional Euclidean space. A smooth acceleration profile can be generated by linearly combining the seven control points. To design the whole acceleration profile, an additional segment with the opposite structure can be constructed in the same Bezier form.

$$a_B(t) = \begin{cases} a_{B,1}(t), & 0 \leq t < T_h \\ a_{B,2}(t - T_h), & T_h \leq t \leq T \end{cases} \quad (3)$$

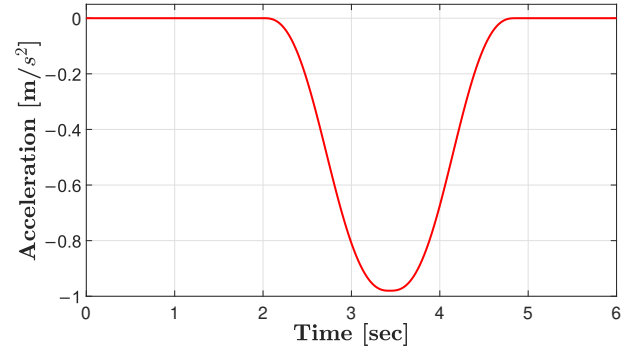


Fig. 7. Acceleration of optimal velocity profile considering human factor.

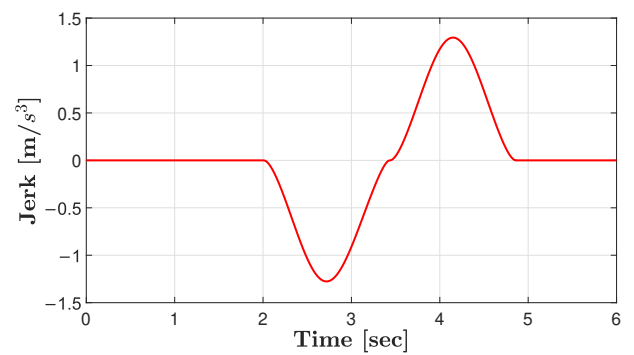


Fig. 8. Jerk of optimal velocity profile considering human factor.

where a_B is a whole acceleration profile constructed by concatenating two Bezier curves in time. T_h is the middle point of the total profile duration. The first segment $a_{B,1}$ is defined in the time interval $[0, T_h]$, and the second segment $a_{B,2}$ is applied in the time interval $[T_h, T]$. To ensure temporal continuity at the junction point T_h , duplicate control points are removed during the generation of the Bezier curve.

By integrating the resulting acceleration profile a_B over time from the initial velocity to zero, a smooth velocity profile can be obtained for the autonomous electric wheelchair.

$$v_B(t) = v_0 + \int_0^t a_B(t) dt \quad (4)$$

where v_0 is the initial velocity, v_B denotes the final velocity profile obtained by integrating the whole acceleration profile a_B over time. Fig. 1 shows the velocity profiles corresponding to five deceleration modes used in the HRI experiment. In each mode, the wheelchair decelerates from an initial velocity of 5 km/h to 0 km/h, and it can be observed that higher deceleration values result in shorter times to reach zero velocity. Figs. 2, 3, and 4 show the corresponding distance, deceleration, and jerk profiles for each mode.

3.2 Optimal Velocity Profile from Human Factor

Based on the designed velocity profiles, the deceleration driving evaluation was conducted with 20 young adult participants. Table 1 shows the results of the de-

Table 1. Results of the deceleration driving evaluation.

Mode	Count	Rank
Mode 1 (-0.2 m/s ²)	3	3
Mode 2 (-0.4 m/s ²)	8	1
Mode 3 (-0.6 m/s ²)	7	2
Mode 4 (-0.8 m/s ²)	1	4
Mode 5 (-1.0 m/s ²)	1	4

celeration driving evaluation experiment. Among the 20 participants, Mode 2 (-0.4 m/s²) was selected as the most preferred mode (Best) by eight participants, making it the most favored mode. In contrast, Mode 4 (-0.8 m/s²) and Mode 5 (-1.0 m/s²) were each selected by only one participant, indicating the lowest preference. Based on these results, the average deceleration preferred by the young adult can be derived as follows:

$$\bar{a} = \frac{1}{N} \sum_{j=1}^5 c_j \cdot a_j \quad \text{where } a_j \in \mathcal{A} \quad (5)$$

where $\bar{a}$ is the average deceleration. N is the number of participants in the experiment. j is five deceleration mode index, and c_j is the number of counts selected as the most preferred by the participants. This reflects the user's needs from HRI experiments, and the optimal deceleration value preferred by young adults is found to be -0.49 m/s². This value can be utilized to design an optimal velocity profile for personalized control in personal mobility service robots.

$$v_B^*(t) = v_0 + \int_0^t a_B(t; \bar{a}) dt \quad (6)$$

where v_B^* is the optimal velocity profile, and $a_B(t; \bar{a})$ is the deceleration based on the Bezier curve by the experimentally derived average deceleration $\bar{a}$. This profile is generated based on the results of the HRI experiment and represents a human factor-informed velocity planning strategy that can be applied to personal mobility service robots to reflect user needs. The optimal velocity profile constructed considering the human factor-based Bezier curve is shown in Fig. 5. Figs. 6, 7, and 8 show the corresponding distance, acceleration, and jerk profiles derived from the optimal velocity profile.

4. OPTIMAL LONGITUDINAL CONTROL CONSIDERING HUMAN FACTOR

To track the optimal velocity profile informed by human factors, the personal mobility service robot is controlled using model-based optimal control techniques. Among them, the autonomous electric wheelchair is modeled as a Two-Wheeled Mobile Robot (TWMR) based on its kinematic and dynamic characteristics [15]. It features two independent driving wheels and two training wheels, allowing for in-place rotation. Fig. 9 shows

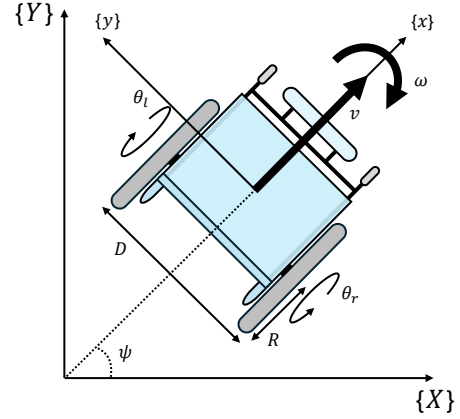


Fig. 9. Autonomous electric wheelchair model.

the structural model and parameters of the autonomous electric wheelchair.

4.1 Autonomous Electric Wheelchair Modeling

The autonomous electric wheelchair is modeled as a differential-drive system, enabling regulation of translational and rotational motion through independent torque inputs applied to the left and right wheels. The kinematic and dynamic equations are derived based on the Newton-Euler formulation, considering straight motion, rotational motion, and the influence of input torques [15]. Accordingly, the continuous-time state-space equation is represented by integrating the system's translational and rotational dynamics with its torque transmission characteristics:

$$\begin{bmatrix} \dot{v} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} \frac{-2B_r}{mR^2+2J_w} & 0 \\ 0 & \frac{-2DB_r}{2JR+D^2J_w} \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} + \begin{bmatrix} \frac{R}{mR^2+2J_w} & \frac{R}{mR^2+2J_w} \\ \frac{-R}{2JR+D^2J_w} & \frac{R}{2JR+D^2J_w} \end{bmatrix} \begin{bmatrix} \tau_r \\ \tau_l \end{bmatrix} \quad (7)$$

where v , $\dot{v}$ are linear velocity and acceleration of the wheelchair, ω , $\dot{\omega}$ are angular velocity and acceleration about the yaw axis. τ_r and τ_l are the input torques applied to the right and left wheels. R is the radius of the driving wheels, D is half the wheelbase, and m denotes the total mass of the wheelchair. The parameter J is the moment of inertia of the wheelchair body, J_w is the moment of inertia of the driving wheel, and B_r is the viscous friction coefficient of the wheels. To compute an optimal control strategy, the continuous-time model is discretized using the Zero-Order Hold (ZOH) method. This discretization allows the computation of control inputs at each sampling time.

4.2 Optimal Stop Control with Error Compensation

To track the human factor-based optimal velocity profile, both longitudinal and lateral control strategies are applied to the autonomous electric wheelchair. The control inputs are defined as:



Fig. 10. Autonomous electric wheelchair robot platform.

$$v^d = v_B^*, \quad \omega^d = 0 \quad (8)$$

where v^d is the desired linear velocity for longitudinal control, and ω^d is the desired angular velocity for lateral control. The optimal velocity profile v_B^* , derived from the human factor-based HRI experiment, is employed as the longitudinal velocity desired for the autonomous electric wheelchair. To achieve this, a state-feedback controller is implemented as:

$$\mathbf{u}_{LQR}(\mathbf{k}) = \mathbf{K}_{LQR} (\mathbf{x}^d(\mathbf{k}) - \mathbf{x}(\mathbf{k})) \quad (9)$$

where $\mathbf{x}^d$ is the desired state vector for tracking $\mathbf{x}$ is the output states of the autonomous electric wheelchair. $\mathbf{k}$ is time step. A Linear-Quadratic Regulator (LQR) is utilized for velocity profile tracking, aim to compute the state-feedback optimal control gain $\mathbf{K}_{LQR}$ that minimizes the following quadratic cost function [16]:

$$\mathcal{J} = \sum_{k=0}^{\infty} (\mathbf{x}(\mathbf{k})^T \mathbf{Q} \mathbf{x}(\mathbf{k}) + \mathbf{u}_{LQR}(\mathbf{k})^T \mathbf{R}_{LQR} \mathbf{u}_{LQR}(\mathbf{k})) \quad (10)$$

where $\mathbf{Q}$ and $\mathbf{R}_{LQR}$ are weighting coefficients for the LQR controller. Solving the discrete-time algebraic Riccati equation (DARE), the autonomous electric wheelchair is capable of tracking the desired reference velocity through the optimal control input $\mathbf{u}_{LQR}$. To enhance tracking performance, a compensator is integrated into the LQR-based controller. The compensator generates an additional control input based on the velocity tracking error between the reference state and the current state. The compensator control law is defined as:

$$\mathbf{u}_{comp}(\mathbf{k}) = \mathbf{K}_{comp} (\mathbf{x}^d(\mathbf{k}) - \mathbf{x}(\mathbf{k})) \quad (11)$$

where $\mathbf{u}_{comp}$ is the compensator output at time step $\mathbf{k}$ and $\mathbf{K}_{comp}$ is the compensator gain that adjusts the error-based correction. Consequently, the final control input, $\mathbf{u}^*$ applied to the autonomous electric wheelchair is the sum of the LQR control and the compensator term:

$$\mathbf{u}^*(\mathbf{k}) = \mathbf{u}_{LQR}(\mathbf{k}) + \mathbf{u}_{comp}(\mathbf{k}). \quad (12)$$

This optimal control strategy allows the wheelchair to accurately track the desired reference velocity while effectively compensating for tracking error.

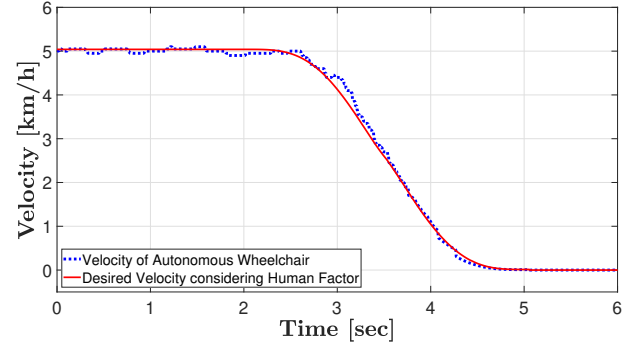


Fig. 11. An autonomous electric wheelchair longitudinal velocity with optimal velocity profile.

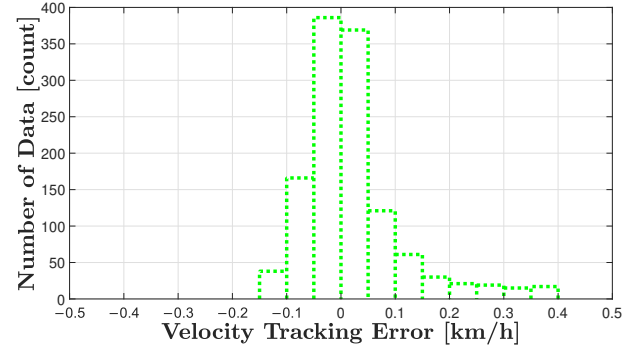


Fig. 12. An autonomous electric wheelchair longitudinal velocity tracking error distribution.

5. EXPERIMENT AND RESULTS

5.1 Experimental Setup

As shown in Fig. 10, an autonomous electric wheelchair robot platform equipped with various sensors and an NVIDIA Jetson AGX Orin Developer Kit embedded PC was used for the experiment. The RPLiDAR S1 LiDAR is mounted on the front and back of the autonomous electric wheelchair robot's armrests to collect environment awareness and positioning information through Simultaneous Localization and Mapping (SLAM)/Navigation. The robot also recognizes obstacles through the Intel Realsense Depth Camera D435i and acquires pose information through the Inertial Measurement Unit VN200. The experiment was conducted at the Gaon Bldg corridor of Pukyong National University under the same environment as the HRI experiment.

5.2 Experiment and Result

Fig. 11 shows the longitudinal velocity tracking performance of the autonomous electric wheelchair during the deceleration scenario. The autonomous electric wheelchair maintains a steady cruising speed of 5 km/h. After 2 seconds, it begins to decelerate according to the optimal velocity profile considering human factor. Wheelchair closely follows the reference velocity profile due to the optimal control strategy integrated with an error-based compensator. Fig. 12 shows the distribution of the velocity tracking error. Most of the error values are centered near zero, and the majority remain within the range of ± 0.2 km/h. This narrow distribution

demonstrates that the proposed optimal stop control with error compensation enables precise velocity tracking and smooth deceleration.

6. CONCLUSIONS

This paper proposed and validated an optimal longitudinal control strategy for autonomous electric wheelchairs, incorporating human factors derived from HRI experiments. A Bezier curve-based velocity profile was designed using the user-preferred deceleration mode, and a model-based optimal control with error compensation was applied to ensure accurate tracking of the desired velocity profile. Experimental results demonstrated that the proposed control system reflected the user's needs, enabling smooth and precise deceleration, contributing to improved driving performance. Future work will focus on extending the human factor model by integrating users' physical motion data, aiming to develop adaptive and personalized control strategies for diverse user needs.

7. ACKNOWLEDGEMENT

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